MR054L ISS Potable Water Quality Monitoring

3.2 Medical Requirements Overview

TABLE 3.2: MEDICAL REQUIREMENTS OVERVIEW

MRID# and Title:	MR054L ISS Potable Water Quality Monitoring
Sponsor:	Medical Operations
Discipline:	Environmental Health System (EHS)
Category:	Medical Requirements (MR)
References:	SSP 50260 ISS Medical Operations Requirements Document
Purpose/Objectives:	Determine and assess crew exposure to chemical contaminants in the potable water supply on the US On-Orbit Segment of ISS based on results from in-flight analyses and ground analysis of archive samples.
Measurement Parameters:	Concentrations of chemical contaminants and values for other water quality parameters measured in samples.
Deliverables:	• Preflight assessment of water chemistry for potable water delivered to the ISS. • In-flight assessment of water chemistry for the potable water supply in the US On-Orbit Segment (USOS) of ISS. • Postflight report assessing the water chemistry of the potable water supply in the USOS based on analysis of archive samples.
Flight Duration:	≥ 30 days
Number of Flights:	Every ISS Increment
Number and Type of Crew Members Required:	One to two crewmembers (CM) are trained in all EHS activities (US Specialist) One EHS CM will perform the in-flight activities
Other Flight Characteristics:	N/A

3.3 Preflight Training

TABLE 3.3: PREFLIGHT TRAINING

Preflight Training Description:	One CM will be trained in all EHS activities (US Specialist).				
Schedule:	**Duration:**		**Schedule (L-months):**	**Flexibility:**	**Personnel Required:**
	Inexperienced Crew: EHS Water Collection:	45 minutes	L-18/15	N/A	Crewmembers/Instructors
	EHS Water Processing	75 minutes	L-12/9		
	EHS Water Sim:	120 minutes	L-9/6		
	Experienced Crew: EHS Water Experienced:	75 minutes	L-15/12	N/A	Crewmembers/Instructors
	EHS Water Sim:	120 minutes	L-9/6		
Ground Support Requirements Hardware/Software:	**Preflight Hardware:**		**Preflight Software:**		**Test Location:**
	EHS Water Kit containing: Water Sample Collection Packet Total Organic Carbon Analyzer (TOCA) Colorimetric Water Quality Monitor Kit (CWQMK) T61p USB to serial adapter T61p Laptop (SSC)		TOCA software CWQMK data transfer spreadsheet		U.S.
Training Facilities:	**Minimum Room Dimensions:**	**Number of Electrical Outlets:**	**Temperature Requirements:**		**Special Lighting:**
	EHS Water Collection & EHS Water Processing: 8' x 10' EHS Water Sim: Bldg 9 SSTF	2 110V	Ambient		N/A
	Hot or Cold Running Water:	**Privacy Requirements:**	**Other:**		
	N/A	Private room free from any distractions	Table & 4-6 chairs 28V Power supply Absorbent towels to collect spillage, if necessary		
Constraints/Special Requirements:	EHS Water Sim takes place in the Space Station Training Facility (SSTF) in Building 9.				
Launch Delay Requirements:	Refresher training will be conducted if currency (18 months) expires or at crewmember request.				
Notes:	Experienced CM – CMs who have had previous training on EHS activities. Inexperienced CM – CMs who have not had previous training on EHS activities.				

3.4 Preflight Activities – No Crew time

TABLE 3.4: PREFLIGHT ACTIVITIES

Preflight Activity Description:	Preflight samples will be collected from all vehicles and/or hardware used to resupply potable water to the USOS of ISS. These samples will be analyzed to verify that the water transferred to ISS meets applicable requirements and is safe for crew consumption.				
Schedule:	**Activity:**	**Duration:**	**Schedule:**	**Flexibility:**	**Personnel Required:**
	Preflight Water Sampling	Vehicle Dependent	Vehicle Dependent	N/A	Vehicle Dependent
Ground Support Requirements Hardware/Software:	**Preflight Hardware:**		**Preflight Software:**	**Test Location:**	
	Ground Servicing Equipment, Approved Sample Containers		None	Vehicle Dependent	
Testing Facilities:	**Minimum Room Dimensions:**	**Number of Electrical Outlets:**	**Temperature Requirements:**	**Special Lighting:**	
	N/A	N/A	Ambient	N/A	
	Hot or Cold Running Water:	**Privacy Requirements:**	**Vibration/Acoustic Isolation:**	**Other:**	
	N/A	N/A	N/A	N/A	
Constraints/Special Requirements:	Detailed logistics (volume and quantity of samples) will be determined by personnel from the JSC Toxicology and Environmental Chemistry (TEC) Laboratory in coordination with visiting vehicle representatives, hardware providers, and the MMOP Water Quality Subgroup.				
Launch Delay Requirements:	Preflight sampling will need to be repeated if the potable water system or water resupply hardware is reprocessed due to a launch delay.				
Notes:	None				
Data Delivery:	Reports from preflight chemical analyses will be provided to JSC TEC personnel and the MMOP Water Quality Subgroup prior to vehicle launch.				

3.5 In-Flight Activities

TABLE 3.5.1: IN-FLIGHT ACTIVITIES

In-Flight Activity Description:	Water samples will be collected from the US Potable Water Dispenser (PWD) for in-flight and ground analysis.				
Schedule:	**Activity:**	**Duration:**	**Schedule:**	**Flexibility:**	**Personnel Required:**
	Water Sample Collection	For PWD: Unstow: 15 minutes Clean Port/Collect Sample: 10 minutes/sample Stow: 15 minutes	See Table D-3 in SSP 50260	N/A	1 Crewmember
	TOCA Analysis	Attended: 10 minutes Unattended: 180 minutes Data Calldown: 5 minutes	See Table D-3 in SSP 50260	N/A	1 Crewmember
	TOCA Waste Water Bag Changeout	Replacement 10 minutes	After every 6 analyses	N/A	1 Crewmember
	TOCA Calibration Check	Attended: 10 minutes Unattended: 180 minutes Data Calldown: 5 minutes	Every 90 days	N/A	1 Crewmember
Procedures:	Procedures are contained within the System Operation Data File (SODF) Med Ops Book				
Constraints / Special Requirements:	Potable Water Sample Collection: • Whenever possible, water chemistry sample collections should be coordinated with water microbiology sample collections. • When chemical samples and microbial samples are collected during the same session, only 15 minutes of unstow time and 15 minutes of stow time is required. • Real-time changes to the sampling schedule and frequency may be made depending on priorities and water systems performance. • Schedule on same day and prior to TOCA analysis. TOCA Operational Constraints: • TOCA analyses should not be scheduled at the same time as T2 exercise. • TOCA analyses should not be scheduled during N2 Leak Checks. • TOCA analyses should not be scheduled during water offloads. Additional water samples may be collected in response to complaints regarding water quality or in contingency situations.				

Photo / TV Requirements:	• Photo documentation is required during TOCA deployment. Photos will be taken at a medium distance to give a reasonable perspective of the deployment location. Additional photo documentation may be requested on an as needed basis.
Cold Stowage Requirements:	N/A
Mission Extension Requirements:	N/A
Landing Wave-Off Requirements:	N/A
Notes:	N/A
Data Delivery:	Any data called down during activities will be logged by the BME and distributed to the Crew Surgeon and to JSC TEC personnel. Downlinked data will be made available to JSC TEC personnel and the flight hardware team personnel upon receipt. A data summary from in-flight analyses will be distributed to stakeholders within 1 week of receipt of downlinked data file. See Table 3.6 Post-flight Activities for archive sample data delivery.

TABLE 3.5.2: IN-FLIGHT HARDWARE

Hardware/Software Name
Total Organic Carbon Analyzer (TOCA)
Colorimetric Water Quality Monitor Kit
Water Sample Collection Packet (Shared with MR051L)
T61p Laptop (SSC)
TOCA USB
USB to serial adapter

3.6 Postflight Activities – No Crew time

TABLE 3.6: POSTFLIGHT ACTIVITIES

Postflight Activity Description:	The archive water samples collected in-flight will be analyzed by the JSC TEC Laboratory.
Constraints/Special Requirements:	Stowage temperatures during transport of the return samples to the TEC Laboratory should be within the range of 2°C to 25°C.
Early Destow / Early Return:	Archive water samples are approved for early destow from IP, US, and COTS vehicles per OB-14-023.
Notes:	SF, SD, OC, or MTLO is responsible for the return of samples to JSC depending on the return vehicle.
Data Delivery:	If the analysis of archive samples indicates an elevation or trend, then JSC TEC will notify the Contingency Action Team, which includes the ISS Increment Lead Crew Surgeon. Analytical results from analysis of routine samples will be reported within 30 days of sample receipt. A final Increment Toxicology report will be posted to the JSC Toxicology Website no later than 3 months after analysis of all Increment samples is completed.

3.7 Summary Schedule

TABLE 3.7: SUMMARY SCHEDULE

ACTIVITY:	DURATION:		SCHEDULE:	PERSONNEL REQUIRED:	CONSTRAINTS:
Preflight Training:					
EHS Water Collection EHS Water Processing	Inexperienced CM:	45 minutes	L-18/15	Crewmember/ Instructor	
	Inexperienced CM	75 minutes	L-12/9		
	Water Sim:	120 minutes	L-9/6	Crewmember/ Instructor	EHS Water Sim takes place in the Space Station Training Facility in Building 9.
EHS Water Experienced	Experienced CM:	75 minutes	L-15/12		
	Water Sim:	120 minutes	L-9/6		
Preflight Activity: – No crew time					
Preflight Water Samples	Vehicle dependent		Vehicle dependent	Vehicle dependent	None
In-Flight Activity:					
Water Sample Collection	For PWD: Unstow: Clean Port/Collect Sample: Stow:	 15 minutes 10 minutes/sample 15 minutes	See Table D-3 in SSP 50260	1 Crewmember	Potable Water Collection • Whenever possible, water chemistry sample collections should be coordinated with water microbiology sample collections. • When chemical samples and microbial samples are collected during the same session, only 15 minutes of unstow time and 15 minutes of stow time is required. • Real-time changes to the sampling schedule and frequency may be made depending on priorities and water systems performance. • Schedule on same day and prior to TOCA analysis.

TABLE 3.7: SUMMARY SCHEDULE (Cont'd)

TOCA Analysis	Crew time 10 minutes Unattended: 180 minutes Data Calldown: 5 minutes	See Table D-3 in SSP 50260	1 Crewmember	• TOCA analyses should not be scheduled at the same time as T2 exercise. • TOCA analyses should not be scheduled during N2 Leak Checks. • TOCA analyses should not be scheduled during water offloads.
TOCA Waste Water Bag Changeout	Replacement: 10 minutes	After every 6 analyses	1 Crewmember	N/A
TOCA Calibration Check	Crew time 10 minutes Unattended: 180 minutes Data Calldown: 5 minutes	Every 90 days	1 Crewmember	- Same as TOCA Analysis
Postflight: - No crew time				
Destow and Return of Samples to JSC (No crew time)	Vehicle dependent	Vehicle dependent	JSC Personnel	• Returned water samples should be maintained between 2ºC - 25ºC and temperatures shall be monitored during storage and transport. • Early destow of water samples and return to JSC is requested.
Postflight Debrief:				
Debrief	No extra time	~R+30 days	Crewmembers/ JSC TEC Personnel	Included as part of the Med Ops overall debrief.